

Summary

Aspiring Backend Engineer with a strong foundation in software development and data analytics, currently pursuing a master’s in computer science. Hands-on experience in network security, backend APIs, and data visualizations. Passionate about solving real-world problems through clean code and scalable systems.

Education

University at Buffalo, Buffalo	Dec 2025
Master of Science in Computer Science	
PES University, Bangalore, India	May 2024
Bachelor of Engineering in Computer Science	

Skills

Languages: Python, C, C++, Java, R, Lua, Go
Web: HTML, CSS, JavaScript, React, Angular, TypeScript, React, NodeJS, Express, Tailwind CSS, Vite
Database: MySQL, PostgreSQL, MongoDB
Tools: Git, Docker, Kubernetes, AWS, Burp Suite, Wireshark
Frameworks: Flask, Django, REST APIs, Spring, MITRE, NIST, CISA

Work Experience

Software Engineer Intern	Dec 2023 – Jun 2024
SecPod Technologies, India	

- Developed detection scripts for emerging cybersecurity vulnerabilities, aligned with CISA advisories and classified using the MITRE ATT&CK framework, in a Kali Linux environment.
- Assessed vulnerabilities and implemented mitigation strategies based on NIST guidelines, using tools like Nessus, OpenVAS, and Qualys to integrate scan results into remediation logic.
- Released secure patches, performed rigorous testing, and optimized detection accuracy—reducing false positives by 20%.
- Enhanced network security through IP routing solutions and collaborated closely with the team during daily stand-ups and sprint planning.

Research Aide - Desktop & Web Application Developer	Aug 2025 – Present
School of Architecture, University at Buffalo	

- Built a full-stack web app with React.js, Vite, Tailwind CSS, Node.js/Python to visualize architectural datasets combining CSV parameters and multi-view images.
- Integrated Recharts, Plotly.js, D3.js for interactive 2D/3D scatter plots, bar charts, histograms, and box plots with dynamic parameter selection.
- Developed features such as condition-based filtering and comparison views, which facilitated more efficient architectural research and data analysis
- Employed Plotly.js for rich data visualization and built reusable React components, resulting in a clean UI/UX that enhanced user experience and project scalability

Projects

- Concurrent TCP Chat System** – C++, POSIX Sockets, select()
- Engineered a robust multi-client chat server using POSIX-compliant TCP sockets and the select() system call for event-driven I/O multiplexing without threads.
 - Handled low-level socket setup, client connection management, and non-blocking command parsing to enable commands like SEND, BROADCAST, BLOCK, and LOGOUT.
- 16-bit Custom Processor Design** – Verilog, Vivado, Basys 3 FPGA
- Designed a modular 16-bit RISC processor with a custom datapath and control logic, supporting instructions such as ADD, SUB, LW, SW, and BEQ.
 - Simulated and synthesized the processor on a Basys 3 FPGA, verifying execution and control flow through testbenches and on-board LED outputs.
- Facial Description-Based Suspect Detection** – Python, PyTorch, TensorFlow, Keras
- Created a facial description-based suspect identification system by integrating descriptions with a knowledge graph to enhance law enforcement capabilities.